

# Scope Management Plan

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## Project Overview:

The project focuses on developing a centralized online food delivery system for Karachi University students. It connects students, canteen owners, and delivery riders through an easy-to-use platform. Students can browse menus, place orders, and receive food within 30–35 minutes without needing to travel to distant canteens. The system reduces student travel costs, saves time, and improves convenience while helping canteens expand their reach. It also ensures faster order processing with fewer errors and aims to achieve high student satisfaction by providing reliable, affordable, and efficient service.

## Scope Management Approach

### (Project Management Body of Knowledge Based)

#### 1. Scope Definition

The project scope will be defined through clear documentation of objectives, deliverables, features, and boundaries. Key stakeholders (students, canteen owners, and IT team) will be consulted through meetings and requirement-gathering sessions to identify needs and expectations. A detailed scope statement, Work Breakdown Structure (WBS), and project charter will be created to outline what is included and excluded from the project.

#### 2. Scope Verification

Scope verification will be carried out through formal reviews and approvals from stakeholders at key milestones. Deliverables will be matched against documented requirements to ensure alignment. Prototypes, demos, and pilot testing will be used to confirm that the system meets user expectations. Final acceptance will require a sign-off from the project sponsor and relevant stakeholders after validation.

#### 3. Scope Control & Change Management

Scope control will be maintained by regularly monitoring progress against the defined scope. Any requested changes will go through a formal Change Control Process, which includes: submitting a change request, assessing its impact on cost, time, and quality, and obtaining approval from the Project Supervisor (or project manager in smaller cases). Approved changes will be documented in the change log, and project plans (schedule, budget, WBS) will be updated accordingly. Regular communication with stakeholders will ensure transparency and prevent scope creep.

# **Roles and Responsibilities**

## **1. Project Manager**

- Define and manage project scope.
- Approve/reject changes.
- Ensure timely delivery within budget and quality standards.
- Communication across teams.

## **2. Business Analyst**

- Gather, analyze, and document requirements.
- Translate business needs into technical specifications.
- A Coordinator between stakeholders and the development team.

## **3. Change Control Board (CCB)**

- Review and evaluate requested scope changes.
- Approve/reject changes based on impact.
- Ensure alignment with project objectives and university policies.

## **4. Developers (Frontend & Backend)**

- Design and implement application features.
- Build APIs, integrate a database, and develop a user interface.
- Ensure performance, security, and scalability of the system.

## **5. UI/UX Designer**

- Create user-friendly and responsive designs.
- Develop blueprints, models, sketches, and design layouts.
- Ensure accessibility and ease of use for students and staff.

## **6. Database Administrator (DBA)**

- Design and maintain database structure.
- Ensure data security, backup, and recovery.
- Optimize queries for system performance.

## **7. Quality Assurance (QA) / Testers**

- Develop test plans and test cases.
- Conduct functional, performance, and usability testing.
- Identify and report bugs, verify fixes before release.

## **8. System Administrator / IT Support**

- Manage hosting, deployment, and server configurations.
- Monitor system uptime and performance.
- Provide technical support for issues.

## **9. Canteen Staff**

- Process and prepare received orders.
- Communicate order status updates to the system through Whatsapp.

## **10. Delivery Personnel**

- Collect orders from the canteen.
- Deliver food to designated student/staff locations.

## **11. Students & Staff (End Users)**

- Browse the menu and place orders.
- Provide feedback on services.
- Report issues or suggest improvements.

## **12. University Administration**

- Oversee system adoption across departments.
- Monitor performance and policies with university guidelines.

## **Scope Definition Process:**

**1. Identify Stakeholders** – Engage students, canteen staff, and university administration to understand their needs and expectations.

**2. Collect Requirements** – Conduct discussions and feedback sessions with students and canteens to capture needs such as menu browsing, cart management, secure login/guest access, and WhatsApp-based order notifications.

**3. Define Project Objectives** – Provide a centralized digital platform for students to order food conveniently, reduce travel and waiting time, and improve canteen order handling efficiency.

**4. Develop Scope Statement** – Document the key features: student-facing interface (menu display, search, cart, order placement), secure login with guest option, admin dashboard for managing canteens and menus, cash-on-delivery payment option, and WhatsApp order notifications for canteens.

**5. Create Work Breakdown Structure (WBS)** – Break down the project into manageable tasks (frontend UI, backend system, database design, testing, deployment).

**6. Review and Approval** – Share the scope document with stakeholders (students, canteen staff, admin) and finalize scope before development begins.

## **WBS Development Method**

The WBS for the KU Campus Bite project was developed using a top-down decomposition approach. The project phases were identified in alignment with the approved scope and then broken down into deliverables and manageable work packages. The project team, under the project manager's guidance, prepared the WBS collaboratively to ensure all scope elements were captured and properly assigned.

## **Scope Verification Process:**

To ensure that completed deliverables meet the project requirements, the following verification criteria and methods will be applied:

### **Verification Criteria**

- The system supports both secure student login and guest access.
- Menus from all participating canteens are displayed correctly with categories and prices.
- Students can browse, add items to cart, and place orders successfully.
- Order details are sent accurately to respective canteens via WhatsApp notifications.
- Food delivery time averages 30–35 minutes (including preparation + delivery), compared to 45–60 minutes with physical travel.
- Basic security measures are implemented (e.g., password encryption, role-based access for admin).

### **Verification Methods**

- **Requirement Traceability Matrix (RTM):** Ensure each requirement is covered by a deliverable.
- **User Acceptance Testing (UAT):** Select a group of students and canteen staff to test features and give feedback.
- **System Testing & Validation:** Verify login/guest access, order placement, WhatsApp notification flow, and admin dashboard functions.
- **Demonstrations & Prototypes:** Present working modules (menu display, ordering process, admin management) to stakeholders.
- **Formal Sign-Off:** Obtain approval from stakeholders that the system meets the agreed scope.

## Scope Change Control Process

Scope changes will be managed through a formal process to ensure the project remains on time, within budget, and aligned with objectives. Any request to modify scope must follow these steps:

- 1. Submission of Change Request** – Any team member or stakeholder may propose a change by documenting its title, description, justification, and potential impact.
- 2. Initial Review** – The Project Manager records the request in the Change Log and reviews it for completeness.
- 3. Feasibility Review** – The development team evaluates technical feasibility, resource needs, and risks.
- 4. Impact Assessment** – The Project Manager, with input from the team, assesses impact on cost, schedule, and quality.
- 5. Decision** – The project team decides to approve, reject, or defer the change, based on alignment with project objectives and available resources.
- 6. Documentation & Communication** – Approved changes are added to the scope statement, WBS, and project schedule. Updates are shared with all stakeholders.

## Tools and Techniques

The following tools and methods will be used to manage project scope and scope changes:

- 1. Documentation Tools** – MS Word and Google Docs will be used to prepare and update project documents (charter, scope statement, requirements, etc.).
- 2. Collaboration Tools** – WhatsApp and Google Meet will be used for quick communication and discussions among team members and stakeholders.
- 3. Change Tracking** – A Change Log (maintained in Google Sheets) will be used to record scope change requests, status, and decisions.
- 4. Version Control** – GitHub will be used to manage code changes, ensuring that scope-related modifications are properly tracked and implemented.

## Scope Management Metrics:

The project scope will be monitored using the following metrics:

1. **Requirements Implementation Rate**– Percentage of functional and non-functional requirements implemented compared to the approved baseline.
2. **Change Requests** – Number of scope changes after baseline approval.
3. **Deliverables Completion**– Percentage of WBS deliverables completed on time.
4. **Stakeholder Feedback** – Satisfaction level with final deliverables.